

MACHINE LEARNING FUNDAMENTALS SERIES

The Confusion Matrix

Classification Performance Evaluation and Misclassification Analysis

Based on StatQuest with Josh Starmer — Machine Learning Fundamentals

1 Introduction and Motivation

When evaluating machine learning classification algorithms—such as Logistic Regression, K -Nearest Neighbors (KNN), or Random Forests—predicting clinical outcomes like heart disease based on patient measurements (e.g., chest pain, blood circulation, weight, blocked arteries) requires detailed performance metrics.

Rather than relying solely on a single aggregate metric like overall accuracy, a **Confusion Matrix** provides a comprehensive breakdown showing what the machine learning algorithm predicted versus the known truth (actual class labels).

2 Structure of a 2×2 Binary Confusion Matrix

In binary classification (e.g., predicting *Heart Disease (+)* vs. *No Heart Disease (-)*):

- **Rows:** Represent the **Predicted Class** output by the machine learning model.
- **Columns:** Represent the **Actual Class** (Known Truth / Gold Standard).

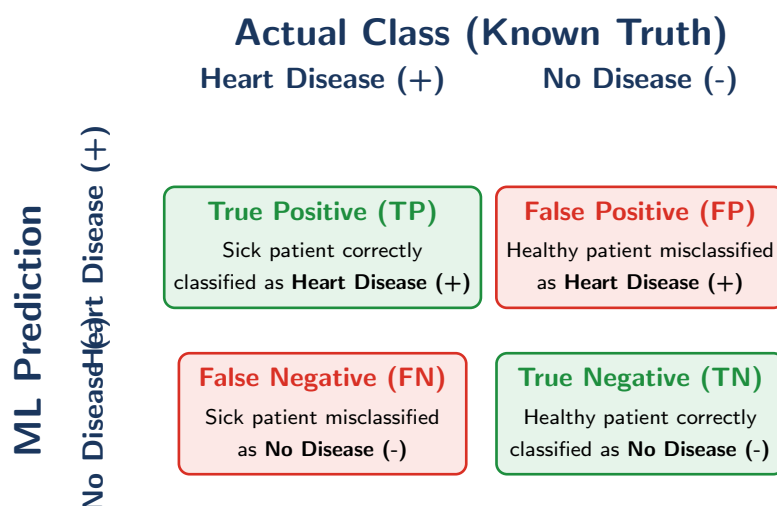


Figure 1: Diagrammatic Illustration of a 2×2 Binary Confusion Matrix. Green cells along the main diagonal represent correct predictions; red cells off-diagonal represent classification errors.

2.1 Matrix Breakdown

- **True Positives (TP):** Top-left corner. Patients who actually have heart disease and were correctly identified by the algorithm.
- **True Negatives (TN):** Bottom-right corner. Patients who do not have heart disease and were correctly identified by the algorithm.

- **False Positives (FP):** Top-right corner. Patients who do not have heart disease but were incorrectly classified as having it.
- **False Negatives (FN):** Bottom-left corner. Patients who actually have heart disease but were incorrectly classified as healthy.
- **The Diagonal Rule:** Numbers along the main green diagonal represent correct predictions. Numbers in off-diagonal red boxes represent misclassifications.

3 Model Performance Comparison

Using testing data, confusion matrices allow direct side-by-side performance comparisons across competing algorithms.

3.1 Random Forest vs. K -Nearest Neighbors (KNN)

On clinical test data for heart disease prediction:

Prediction \ Actual	Random Forest		K-Nearest Neighbors	
	Disease (+)	No Disease (-)	Disease (+)	No Disease (-)
Heart Disease (+)	142 (TP)	22 (FP)	107 (TP)	? (FP)
No Disease (-)	29 (FN)	110 (TN)	? (FN)	79 (TN)

Table 1: Comparison of Testing Data Confusion Matrices for Random Forest vs. K -Nearest Neighbors.

- **Random Forest:** Achieved **142 True Positives** and **110 True Negatives**, with 29 False Negatives and 22 False Positives.
- **K-Nearest Neighbors (KNN):** Achieved only **107 True Positives** and **79 True Negatives**.
- **Decision:** Random Forest significantly outperforms KNN on both positive and negative class predictions.

3.2 Logistic Regression & Advanced Metrics

When Logistic Regression is applied, its confusion matrix is very close to Random Forest. Choosing between two close models motivates more advanced evaluation metrics:

- **Sensitivity (Recall):** $\frac{TP}{TP+FN}$ — Measures the proportion of actual sick patients correctly identified.
- **Specificity:** $\frac{TN}{TN+FP}$ — Measures the proportion of actual healthy patients correctly identified.
- **ROC and AUC:** Receiver Operating Characteristic curves and Area Under Curve analysis.

4 Multiclass Confusion Matrices ($N \times N$)

The dimensions of a confusion matrix scale directly with the number of target classes:

- **3 Categories** (e.g., predicting favorite movie among 3 options): Generates a 3×3 matrix.
- **N Categories:** Generates an $N \times N$ matrix with N rows and N columns.

- **Universal Rule:** The main diagonal always contains correct classifications, while all off-diagonal cells contain misclassification counts.